EPSR-test-ex83 dockato: water onto DMSO (docked molecule with ROT groups)

Program(s): dockato | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

`dockato` [1] builds larger molecules from `.ato` building blocks: copies of the first molecule of a docking file are bonded onto free sites of a docked file, removing one atom of each to form the new bond. Here the docked molecule already has rotational groups, which must be shifted when a new group is inserted.

Code exercised

Fortran (original, fortran_src/):

- `dockato.f`: `dockato` (driver, parameters from command line or stdin), `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`
- `dockato.f`: `test_overlap`, `bond_molecule` (ROT/DIH groups), `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`
- `ato_manipulation_rev4.f`: `get_direct_bonds`, `get_rotgrp`, `centreofmass`, `ftype`; `ato_read_and_save.f`: `moloutc`

Rust (port, rust/epsr-core/src/):

- `tools/dockato.rs`: `run`, `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`, `test_overlap`
- `tools/dockato.rs`: `bond_molecule`, `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`, `get_direct_bonds`, `get_rotgrp`
- `ato.rs`: `State::moloutc` (ROT/DIH groups), `State::ftype`; `rng.rs`: `Ran1::ran1`

Physical system

Building blocks of EPSR-test: `DMSO.ato` (one DMSO, (CH3)2SO, S-C 1.82 Å, C-H 1.10 Å, two methyl ROT groups) and a water molecule. Atom `atomd` of the new molecule is placed on `atomv`, the molecule is rotated so that its `atomdr` neighbour lies on the `atomvr` neighbour of `atomv` (equal bond lengths required), an optional second (dihedral) rotation about the new bond uses `atomdrr`/`atomvrr` (random angle in [dminrr, dmaxrr] from `ran1`, the Park-Miller generator with Bays-Durham shuffle [2,3], or a fixed dihedral that creates a DIH group when dminrr = dmaxrr), overlaps are tested with a distance factor, `atomd` and `atomvr` are removed and the new rotational group, bonds and atom types are updated. The water H1 is docked onto a DMSO C and its O1 onto the C-H hydrogen, forming a C-O bond (a methoxy-like substitution of a methyl H); the existing DMSO ROT groups are shifted and a new ROT or DIH group is inserted.

Inputs

`WaterOH110.ato` = `WaterMolecule.ato` with the O-H bond entries set to 1.10 Å (coordinates unchanged), because the two site pairs need equal bond lengths. Run 1: site answers `C H H`, 30-90°, factor 1.0 from `dmso_random.stdin`, 3 molecules; run 2: fixed dihedral 180° referenced to S, overlaps allowed, 4 molecules; run 3: no second rotation, factor 0.9, 6 molecules.

How the accuracy was verified

`commands.txt` (3 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Each run writes one new `.ato` file (coordinates, bonds, rotational groups, atom types and the random-number state), compared byte for byte with the stored Fortran output. Outputs: `dmsowater.ato` 13 atoms (3 dockings) with 5 ROT groups; `dmsodih.ato` 14 atoms with 2 ROT + 4 DIH groups; `dmsonorot.ato` 16 atoms (6 dockings) with 2 ROT + 6 DIH groups.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] S. K. Park and K. W. Miller, "Random number generators: good ones are hard to find", Commun. ACM 31, 1192-1201 (1988).
[3] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, Numerical Recipes in Fortran 77, 2nd ed., Cambridge University Press (1992).
[4] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[5] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.